

Embedded visualizations in crane operation user interfaces for real-time assistance

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ARTICLE INFO

Keywords:

Crane
Safety
Embedded visualizations
User Interface
Augmented reality

ABSTRACT

While crane operations are becoming increasingly complex, challenges remain in creating user interfaces (UIs) that effectively support real-time decision-making and situational awareness. This paper presents a comprehensive review and evaluation of visualizations used in crane operations, focusing on user interfaces designed to enhance operator performance and safety during lifts. Through a systematic review of existing literature, this paper synthesizes key trends, design principles, and technologies in crane UIs, with a particular emphasis on the role of embedded visualizations. A tailored UI evaluation model is developed, drawing from principles in human-computer interaction and Augmented Reality (AR) design realms, to assess the efficacy of these systems in real-time operations. This review also identifies key research gaps, including the need for empirical testing of individual visualizations and comprehensive UI configurations to better understand their impact on operator performance. Overall, this paper makes valuable contributions to the field by laying the groundwork for improving both the safety and productivity of crane operations through more effective, user-centered UIs.

1. Introduction

Cranes are indispensable on construction sites owing to their ability to expedite the movement of heavy objects around construction sites [1]. This prevalence, however, is matched by a significant involvement in construction related accidents worldwide. Between 2011 and 2017, the United States alone recorded 297 fatalities due to crane-related accidents [2]. A global perspective from Milazzo et al. [3] reveals a broader concern, with 937 crane-related accidents recorded worldwide from 2011 to 2015. More recent data from the Occupational Safety and Health Administration (OSHA) shows that between 2017 and 2024, 850 crane-related accidents were recorded in the United States with 287 (33.8 %) of them involving at least one fatality [4]. Notably, it is likely that many crane-related accidents remain unreported, suggesting that this issue is far larger, particularly in countries that do not have robust incident reporting and record systems [5,6]. Furthermore, modern-day crane operations are becoming increasingly complex due to upward trends in industries like modular construction [7,8]. Many of these operations involve critical lifts, which are defined as those that exceed 75 % of a crane's rated capacity or require multiple cranes due to excessively large loads [9]. Such lifts necessitate more stringent planning and

safety measures to prevent accidents as potential losses are more substantial.

In response to this, much research attention has already been dedicated to improving crane lift planning procedures to pre-emptively minimize potential hazards faced by crane operators [10–12]. Despite these efforts, construction sites are notoriously dynamic environments, demanding constant vigilance from crane operators to navigate the ever-changing surroundings and avoid costly errors. Unfortunately, the nature of the job, involving repetitive tasks and long working hours, means even the most competent crane operators are prone to potential distractions and fatigue [13], thereby compromising situational awareness and decision-making ability. This issue is further exacerbated by current global skilled labor shortages in countries such as Japan [14], and Hong Kong [15]. Other developed countries such as the United States [16] and New Zealand [17] have also cited specific shortages in competent crane operators. Consequently, companies are being pressured to fill vacancies left by retiring senior operators by hiring less experienced workers or extending the working hours of current employees, possibly increasing the likelihood of errors. A study of 75 crane accidents by King [18] showed that 32 (42.7 %) occurred where the operator did not fulfill their responsibilities. In fact, several studies have determined that human

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<https://doi.org/10.1016/j.autcon.2025.106078>

Received 1 November 2024; Received in revised form 13 February 2025; Accepted 18 February 2025

Available online 28 February 2025

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error is the primary factor in about 80 % of all construction accidents [19–21]. Moreover, a recent expert study revealed that current crane operations suffer from a lack of contemporary and updated strategies or policies. Consequently, accidents related to these operations are often observed to be the outcome of human errors [22]. Overall, this indicates a need for further development of real-time measures to assist crane operators during the construction phase, as they are primarily responsible for the safety of a lift job [23].

A significant branch of research in this area follows the use of Real-Time Location Systems (RTLS) such as Radio Frequency Identification (RFID), Ultra-Wideband (UWB), and computer vision technologies. These systems can not only capture crane and load motions with minimal latency but also are able to capture data of the surrounding environment, providing invaluable data to be processed into feedback for operators in real-time. Multiple studies have highlighted the potential benefits to safety and performance when offering additional feedback to crane operators through a variety of sensory channels, including dynamic visual and audio cues via advanced user-interfaces (UI) [24–26], as well as haptic feedback through control stick vibrations [27]. These studies have highlighted that supplementing operator situational awareness in a more dynamic and interactive manner helps them to make better decisions leading to fewer errors. These advancements mark a substantial departure from traditional Load Moment Indication (LMI) systems in cranes which rely on simple graphics and numerical values.

From an alternative perspective, Augmented Reality (AR) technologies have gained increasing popularity as tools for providing feedback to users, aiming to enhance performance by overlaying digital information (i.e., visualizations) onto a user's real-world view [28]. The distinctive strength of AR lies in its capability to map the physical environment and overlay context-aware visualizations onto referent objects or locations that assist in discerning spatial relationships and decision-making [29]. Notably, advanced driver assistance systems (ADAS) in modern automobiles leverage AR to provide drivers with real-time visualizations and guidance through a heads-up display (HUD) on the windshield, thereby improving situational awareness and reducing accident risks [30]. Although the specific implementations differ, the underlying principles of using AR to enhance situational awareness and support real-time decision-making in crane operations remains underexplored. This limited number of studies can be attributed to current challenges in AR development including technological limitations [31] such as registration accuracy, restricted Field-of-View (FOV), and high latency, which are particularly undesirable in high-risk lift activities. From another perspective, other social and economic limitations [32] also hinder AR adoption. For instance, operators require extensive training to effectively use AR systems, and their willingness to adopt new technologies can vary especially among operators who are not technologically savvy. Additionally, the high initial and maintenance costs of these systems can be difficult to justify for industry supporters, given the uncertain returns on investment. Nevertheless, some researchers have overcome these limitations by demonstrating AR implementations in laboratory-scale crane lift environments [25,33], showing promising results in improving efficiency and reducing mental load. Others have also conducted real-world applications in other heavy machinery such as forklifts [34] and forestry machines [35] to enhance operator visibility. These advancements suggest that AR implementations in cranes are gradually becoming more feasible and are worth further exploration.

The existing body of literature on AR applications in the construction industry highlights the transformative potential of AR technologies in enhancing safety, efficiency, and overall project outcomes. Several reviews have explored the multifaceted applications of AR, emphasizing its contributions to construction safety, stakeholder engagement, and training [36–40]. However, the majority of these studies delve little into design considerations for AR use in heavy equipment in general, much less in cranes. Ultimately, one review felt that it appears premature to provide definitive guidance on the development of AR UIs within the realm of heavy machinery based on the vast range of equipment and

applications [40]. From another perspective, researchers in the field of Immersive Analytics [41] have suggested that designing AR visualizations for a UI in dynamic environments that constantly change exponentially increases the complexity of the task. Given the demands on crane operators to maintain high levels of situational awareness, the development of these UIs must strike a delicate balance between providing the most useful information and not overwhelming the operator. Furthermore, this information needs to be easily perceived, yet not overly intrusive, ensuring that critical site elements such as the load are not obstructed. These challenges highlight the necessity of carefully designing AR UIs tailored to the unique demands of crane operations.

Crane operations typically involve lifting heavy loads to precise locations, often at significant heights, demanding specialized expertise and an acute awareness of safety protocols distinct from those required in other heavy machinery operations. Thus, given the specific nature of crane operations and their pivotal role in construction projects, there is a significant gap in the literature concerning the design and implementation of visual information in AR UIs tailored explicitly for crane operators. This paper seeks to fill this gap by focusing on the unique challenges and requirements of crane operations, contributing to the broader field of AR applications in construction safety and operation.

2. Research methodology

This paper aims to assess the current state of visual information systems for crane operators, develop a tailored UI evaluation framework for real-time crane operations, and propose future research directions based on these findings. The research methodology encompasses a systematic review of relevant scholarly literature and the development of a comprehensive UI evaluation model.

The literature review process focuses on identifying key trends, technologies, and design principles in existing UI systems for crane operators. Additionally, the evaluation model, tailored specifically for real-time crane operations, assesses UI design across several dimensions, such as consistency, clarity, and usability.

The remainder of the paper is structured as follows: First, the remainder of section 2 details the literature search process and the analysis methods including the development of a UI evaluation model. Next, section 3 follows with an analysis and discussion of trends observed in the reviewed literature and the application of the evaluation framework. Next, section 4 outlines future research directions for further studies in this field. Finally, section 5 concludes by summarizing the key contributions of this paper to the field.

2.1. Literature search

This paper employs a systematic four-step literature search methodology. Firstly, it initiates the search by querying relevant papers within the Scopus and Web of Science literature databases. To ensure inclusivity, a comprehensive search string was developed, encompassing various forms of real-time visualization assistance designed for crane operators. For instance, keywords such as “virtual reality”, “simulation”, “teleoperation”, and “remote operation” were included to capture insights in real-time visualizations for crane operators that might have been explored in simulated or remote environments. The specified publication period, spanning from 2000 to 2024, was chosen to focus on recent developments in the field. The full search string is as follows:

TITLE-ABS-KEY ((“crane”) AND (“virtual reality” OR “augmented reality” OR “mixed reality” OR “extended reality” OR “windshield” OR “transparent” OR “see through” OR simulat* OR “real time”) AND (“remote operation” OR “teleoperation” OR operat*) AND (annotat* OR visual* OR “display” OR *interface OR “alert” OR assistan*)).

Note that wildcard operators (i.e., * symbol) are used to account for word variations (e.g., annotat* covers annotation, annotate, annotating,

etc.).

Secondly, the search results were refined through an initial screening of the titles and abstracts. An initial set of exclusion criteria were applied, eliminating studies not centered around crane equipment (e.g., other machinery, crane bird), studies outside of operational processes (e.g., planning, inspection, education, process modeling, equipment identification), and non-English studies. The third step involved a detailed manual screening, prioritizing studies introducing novel visualizations or non-traditional methods of presenting existing information on traditional LMI displays (example in Fig. 1). Papers failing to meet these criteria were excluded, including those that did not present information in real-time, those exclusively focusing on automatic correction/response assistance mechanisms, and those centered on equipment simulation without addressing information delivery. The fourth and final step involves adding additional relevant papers identified through citation searches to ensure a comprehensive review. The flowchart in Fig. 2 summarizes the literature selection process. It is worth noting that studies lacking visual feedback (i.e., limited to haptic/audio feedback) were deliberately excluded from this review. In essence, this systematic review places primary emphasis on exploring advancements in visual information delivery for real-time crane operation procedures.

2.2. Literature analysis

The literature analysis in this review is structured around two key components: a categorization framework and a UI evaluation model. Together, these elements enable a comprehensive examination of the current state of visualizations for real-time crane operations and provide insights into the design considerations driving UI development.

2.2.1. Categorization of reviewed studies

To systematically analyze the selected literature, the studies are categorized by key factors that showcase the different approaches to UI and visualization design for real-time crane operations. These categories include crane type, the information provided to operators, the corresponding visualizations, and their level of embeddedness. Additionally, the location of the visualizations and the test scenarios used for validation are considered. This structured approach enables clearer comparisons across studies, highlighting trends and variations in UI design for crane operators.

2.2.2. UI evaluation model

The literature analysis employs a comprehensive evaluation model that integrates established UI design considerations to evaluate the effectiveness of real-time visualizations for crane operators during

construction operations. Beyond assessing the individual papers, the goal is to provide an insightful examination of the state of the art concerning the incorporation of design considerations in the field of innovative crane operation UIs for real-time assistance.

The evaluation model employed for this literature analysis is developed by drawing upon a synthesis of design rules, principles, and heuristics from diverse sources in closely related fields associated with interface design. Specifically, these included Human Machine Interface (HMI) [43,44], Graphic User Interface (GUI) [45,46], and Augmented Reality (AR) [47,48] design domains. These fields were selected for their relevance in developing effective UIs in dynamic environments, such as crane operations, ensuring the model's robustness and applicability. Drawing from this diversity, the evaluation framework is equipped to address the unique challenges of real-time operational environments.

Each design consideration was meticulously analyzed, with similar ones consolidated to reduce redundancy and create a comprehensive yet concise evaluation model. Consequently, a model containing 19 separate principles was developed. The evaluation model is structured under five key design principle groups: 'Consistency', 'Clarity', 'Organization', 'Usability and Compatibility', and 'Error Tolerance and Recovery'.

- *Consistency* ensures uniformity and stability of components, leveraging the operator's familiarity with recurring design patterns and maintaining stability in the placement and function of interface elements.
- *Clarity* enhances visual comprehension through aesthetically clear and intuitive designs, ensuring that information is context-aware and immediately understandable.
- *Organization* focuses on structuring information in a logical hierarchy, aiding operators in decision-making by grouping related controls and data efficiently.
- *Usability and Compatibility* prioritize interface efficiency, adaptability, and user satisfaction, aiming to reduce operator workload, increase flexibility in data presentation, and simplify interactions.
- *Error Tolerance and Recovery* deals with preventing user errors and providing effective recovery mechanisms, ensuring safe and seamless crane operations by guiding the operator through error rectification.

By categorizing specific principles under these overarching groups, the evaluation model provides a systematic lens for evaluating the effectiveness of real-time crane lift UIs. Table 1 outlines each principle and contextualizes them from a crane operator's perspective, highlighting the specific sources from which each principle was derived. Each principle is designed to address critical UI design factors that contribute to the overall user experience and, consequently, operator safety and performance. Hence, it should be noted that certain principles pertaining to device maintenance, data privacy, device setup and documentation were excluded from the model. While these aspects are undoubtedly crucial in the broader spectrum of UI implementation, their exclusion allows for a more targeted analysis of design principles directly influencing the real-time user experience of crane operators in an active lift job.

To evaluate the fulfillment of these design principles across studies, this paper adopts a binary classification system using the terms 'Observed' and 'Unobserved'. A principle is considered 'Observed' when any of the following criteria are met:

- 1) an analysis of figures featuring UI elements reveals the presence and implementation of the principle (e.g., Error Resolution – error messages are clearly depicted on the UI).
- 2) in-text descriptions explicitly mention the application or consideration of the principle (e.g., Virtual Object Integration – study mentions considerations around visibility and positioning of AR visualizations).

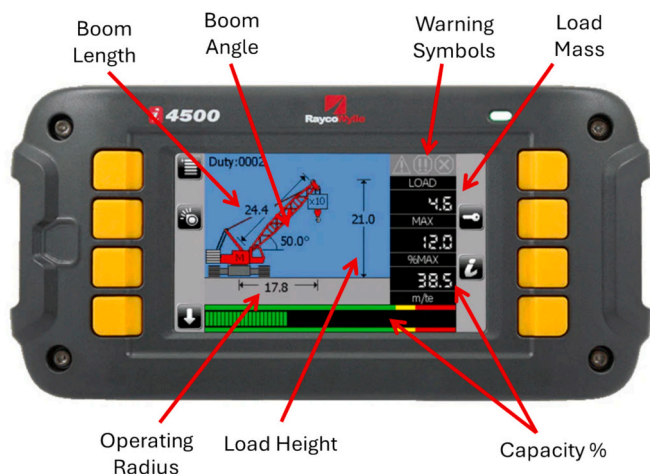


Fig. 1. Typical crane Load Moment Indicator (LMI) system – Modified from [42].

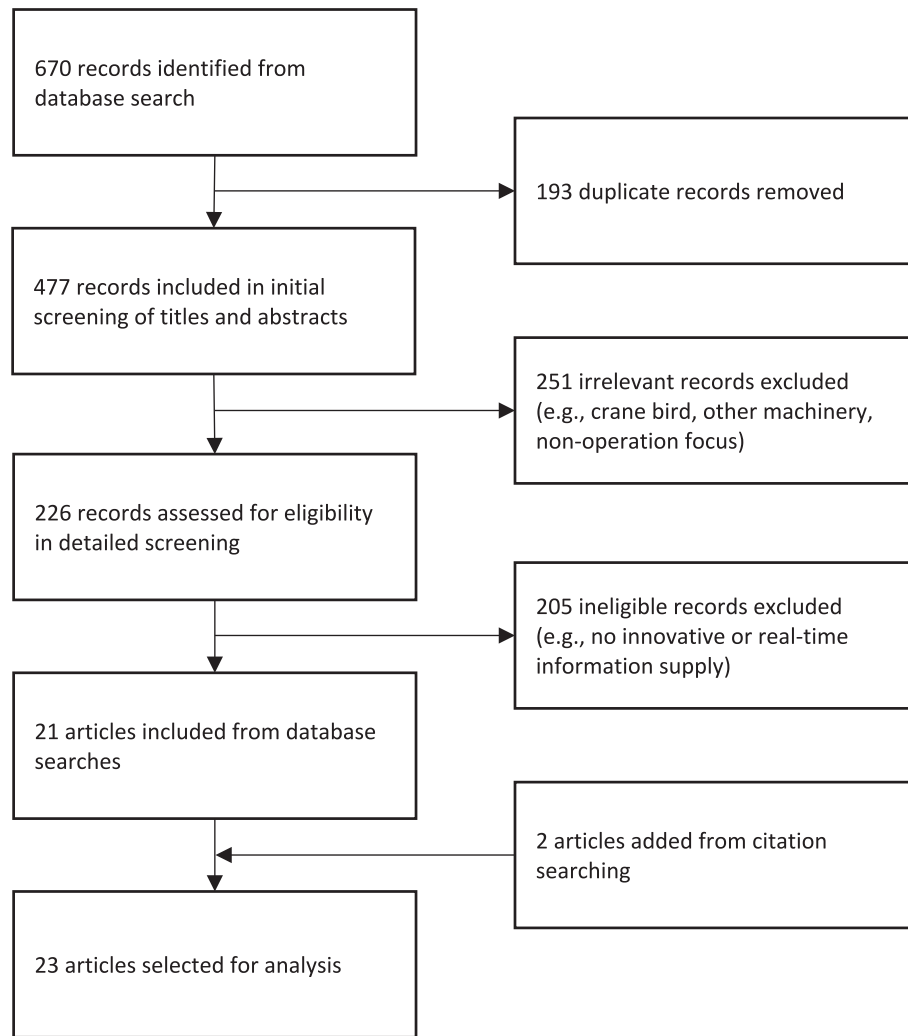


Fig. 2. Literature selection flowchart (as of July 2024).

3) other materials or appendices provide evidence supporting the principle's use (e.g., Feeling of Control – questionnaires to participants ask users to rate if they feel in control).

Conversely, a principle is 'Unobserved' when none of these criteria are satisfied. This binary approach is favored due to the difficulty in directly comparing how each criterion is fulfilled across diverse studies. Each principle can be implemented in various ways, making a more nuanced or tiered evaluation challenging to evaluate objectively. The Observed/Unobserved system provides a flexible yet structured framework, allowing for an insightful examination of UI designs without overcomplicating the comparison process.

3. Results and discussion

3.1. Literature search overview

The analysis covers 23 articles selected for their relevance to innovative methods of real-time information supply to crane operators. Although the search spanned from 2000 onwards, the earliest relevant article dates back to 2011, reflecting the relatively recent emergence of research in this area. During data collection, it was noted that some articles from the same research groups used similar or updated versions of previously developed UIs. In these cases, the UIs were grouped to provide a more cohesive evaluation with further analysis conducted primarily using the most updated study. As a result, the publication list

was condensed to 18 distinct studies. Table 2 summarizes the findings of these studies based on the categorization framework outlined in the methodology.

3.2. Categorical analysis

3.2.1. Crane types

The distribution of crane types across the reviewed studies reflects their prevalence and operational importance in real-world scenarios. Mobile cranes account for 50 % of the reviewed literature, followed by tower cranes at 39 %, and gantry cranes at 11 %. This distribution suggests that research interest is closely aligned with the versatility and operational demands of these crane types, highlighting areas where improvements in safety and efficiency are most critical.

Mobile cranes dominate the literature, representing half of the reviewed studies. Their widespread application across various industries and adaptability to different environments likely contribute to this focus. Mobile cranes are frequently used on construction sites of varying scales, where their ability to move between locations is essential. This mobility presents unique UI design challenges, particularly in adapting to diverse spatial constraints and changing site conditions. For instance, a mobile crane might need to navigate through tight urban construction sites one day and open rural areas the next, requiring UIs that can seamlessly transition between these contexts and provide relevant information accordingly. While mobile cranes often operate at ground level with fewer visible elements compared to tower cranes, UIs must

Table 1

Design principles tailored to the real-time crane operation context.

Principle Group	Design Principle	[43]	[44]	[45]	[46]	[47]	[48]
Consistency	Leveraging Occupational Familiarity: Enhance user experience and affordance by incorporating familiar conventions from crane operation standards, ensuring that the interface aligns with industry expectations.		✓	✓	✓	✓	✓
	Stability of Components: Maintain consistency in the position and functionality of critical UI components, providing operators with a stable and predictable interface during the dynamic phases of crane operations.			✓			
	Uniform Design for Similar Components: Avoid casual differences in the visual design of components with similar functions (and vice versa) without valid reasoning (e.g., innovative conventions), helping operators easily identify and interact with elements related to specific crane functionalities.		✓		✓		
Clarity	Aesthetic and Readable Designs: Create visually appealing designs with appropriately sized components to ensure readability, promoting a visually clear and comfortable interface for crane operators.			✓		✓	
	Context-Aware Designs: Prioritize critical information based on the current context of crane operations, presenting relevant data prominently while minimizing unnecessary details to avoid clutter and cognitive overload.		✓	✓	✓		✓
	Distinct and Intuitive Visual Elements: Utilize graphics and terminology that convey clear meanings to operators, promoting quick and accurate interpretation of information without the need for additional cognitive effort.	✓	✓	✓	✓	✓	✓
Organization	Multiple Views for Complex Elements: Implement multiple view angles to provide crane operators with diverse perspectives in complex lift tasks, aiding in a comprehensive understanding of the operational environment.				✓		
	Use of Color and Graphics as Perception Aids: Leverage color and graphics strategically to enhance the perception of information, facilitating quicker decision-making and reducing cognitive strain.		✓	✓			
	Virtual Object Integration: Ensure seamless integration of virtual elements into the real-world environment, preventing obstruction of crucial information and objects during crane lift operations.					✓	✓
Usability and Compatibility	Structured Information Delivery: Sequence information delivery to signal clear transitions in a manner that crane operators can anticipate, facilitating smooth execution of operations.	✓		✓			
	Logical Grouping of Information and Controls: Arrange information and controls in an accessible and hierarchical manner, establishing clear relationships to enhance comprehension and streamline the operator's decision-making process.		✓	✓	✓	✓	✓
	Manage User Attention: Direct user attention to critical elements through cues and highlighting, ensuring that operators focus on the most relevant information during the various stages of crane lift operations.				✓		
Error Tolerance and Recovery	Automation of Workload: Automate or streamline mental calculations and estimations to reduce cognitive load, allowing crane operators to allocate their mental resources more effectively during operations.		✓				
	Flexible Data Presentation: Provide configurable interface settings to present data in varying formats, accommodating different user preferences, and promoting cognitive flexibility.		✓	✓			
	Simplicity of Controls: Ensure controls are simple, minimizing eye and hand movements for interface interaction, thereby reducing confusion and enhancing operator efficiency and safety during lift operations.	✓		✓	✓	✓	✓
	Feeling of Control: Instill a sense of control in crane operators, ensuring that the interface responds predictably to their actions and provides a feeling of mastery during operations.	✓		✓			
	Collaborative Interface: Design the interface with the collaborative nature of crane lifts in mind, facilitating effective communication and coordination between the operator, signalpersons, riggers, and other cranes to ensure safe and efficient operations.					✓	
	Protection Against Serious Errors: Implement preventive measures in the interface, such as automatic cues and reminders, to safeguard against serious errors that could compromise crane safety.	✓		✓			
	Error Resolution: Account for operator errors by providing clear error messages/directions with actionable feedback, allowing crane operators to rectify mistakes easily and maintain smooth operations.			✓		✓	

prioritize this flexibility and account for intuitive features that allow for easy pre-lift calibration.

Tower cranes are the second most studied, accounting for 39 % of the literature. These cranes play a central role in large-scale construction projects, operating from fixed, elevated positions with extensive reach. This static but elevated operation introduces distinct visibility challenges, as operators must monitor a broad area filled with dynamic elements such as personnel, equipment, and materials, some of which may be outside their direct line of sight. As a result, UI design for tower cranes emphasizes comprehensive situational awareness. The UIs must ensure that key elements relevant to the current task are clearly highlighted, even if they are distant or obscured. Given the crane's height and large operating radius, UIs should help operators manage both immediate and peripheral site activity, ensuring critical elements aren't missed. For example, a tower crane operator might need to lift materials from a crowded construction site to the top of a high-rise building. The UI must provide a clear and comprehensive view of both the immediate area around the crane and the distant drop-off point, ensuring that the operator can safely navigate obstacles and coordinate with ground personnel. The complexity of the construction environment, combined with the tower crane's immobility, calls for UIs that can seamlessly track

and display important information across a wide FOV.

Gantry cranes represent only 11 % of the reviewed studies, reflecting their more specialized use in controlled environments such as ports and industrial yards. These cranes operate in relatively predictable and repetitive conditions, which likely explains the smaller research focus. Unlike mobile and tower cranes, gantry cranes are less dynamic in their movements and are typically used for highly specific tasks like loading and unloading cargo or moving heavy materials over short distances. The predictability of gantry crane operations simplifies UI design in some respects, with less emphasis on managing a broad range of dynamic elements. Instead, UIs for gantry cranes focus on precision, supporting operators in carrying out repetitive tasks efficiently. For instance, in a port setting, a gantry crane operator might repeatedly load containers onto ships. UIs should provide precise control and feedback mechanisms to ensure that each container is accurately placed, minimizing the risk of errors in a highly repetitive task. Since the operational context is more controlled, real-time updates may be less critical, but interfaces still need to ensure accuracy and clear visualization of task-relevant information.

This distribution highlights how the operational context of each crane type may influence UI design. For mobile cranes, adaptability and

Table 2

Literature Result Summary (E – Embedded; NE – Non-Embedded; PV – Primary View; SV – Secondary View). For a detailed illustration and description of PV and SV, refer to Fig. 4 in Section 3.2.4.

Ref.	Year	Machine Type	Supplied Information	Corresponding Vis.	Vis. Embeddedness	Vis. Location	Test Scenario
[49]	2011	Tower Crane	Load delivery zone Obstacles Target load Target load (blind lift)	Colored area (Green) Colored area (Red) Colored area (White) Text (distance behind obstruction)	E E E E	PV PV PV PV	In-cabin operation (virtual)
[26]	2012	Tower Crane	Clearance warning (to crane boom)	Text (Blinking)	NE	SV	Remote operation (laboratory)
[50]	2012	Tower Crane	Boom angle Load height Load weight Maximum slewing radius Slewing angle	Text Text Text Colored area (Red) Text	E E E E E	SV SV SV SV SV	In-cabin operation (in-situ)
[51]	2014	Tower Crane	Lift path (Planned)	Colored line (Yellow)	E	PV	Remote operation (virtual)
[52]	2015	Tower Crane	Load height Load weight Slewing angle Slewing radius	Text Text Text Text	NE NE NE NE	PV PV PV PV	In-cabin operation (virtual)
[33], [25]	2016	Tower Crane	Lift path (Planned) Load delivery zone Load height Clearance distance (to load) Clearance warning (to load) Clearance warning (to load)	Colored line (Yellow) Colored area (Yellow) Text Text Colored text highlight (Red) Warning symbol	E E NE NE NE NE	PV PV SV SV SV PV	Remote operation (laboratory)
[53], [54]	2016	Mobile Crane	Boom angle Boom length Lift path (Planned) Obstacles Clearance warning (to crane boom) Safe operating zone	Text Text Text Colored area (Red) Text (Blinking) Colored area (Green)	NE NE NE E NE E	SV SV SV SV SV SV	In-cabin operation (in-situ)
[55]	2017	Gantry Crane	Clearance warning (to crane hook) Clearance warning (to crane hook)	Colored highlight (Red, Blinking) Colored area (Blue)	NE E	SV PV	In-cabin operation (in-situ)
[56], [57]	2017	Mobile Crane	Boom length Lift path (Executed) Load sway warning Load weight Maximum slewing radius Obstacles Clearance warning (to load) Clearance warning (to load)	Text Colored line (Red) Colored text highlight (Red) Text Text Colored bounding box (Grey) Colored load highlight (Yellow) Colored obstacle highlight (Yellow)	NE E NE NE NE E E E	SV SV SV SV SV SV SV SV	In-cabin operation (in-situ)
[58]	2018	Mobile Crane	Clearance warning (to load) Boom angle Boom length Slewing radius Slewing radius	Colored text highlight (Red) Text value Text value Colored arc (Red) Text value	NE NE NE E NE	SV SV SV SV SV	In-cabin operation (virtual)
[59], [60]	2020	Mobile Crane	Hazard response guidance Clearance distance (to crane) Clearance warning (to crane) Clearance warning (to crane)	Text Text Colored arrows (Red) Colored obstacle highlight (Yellow)	NE NE E E	SV SV SV SV	In-cabin operation (in-situ)
[61], [62]	2020	Mobile Crane	Clearance warning (to crane) Target load Boom angle Boom length Crane balance Lift capacity Load height Load sway Lift capacity warning Clearance warning (to crane) Slewing radius Wind direction	Text Colored area (Red) Text Text Colored directional symbols (Yellow) Colored capacity meter (Yellow) Text Colored directional symbols (Yellow) Colored warning symbol (Yellow) Colored directional symbols (Yellow) Text Colored directional symbols (Yellow)	NE E NE NE NE NE NE NE NE NE NE NE	SV SV PV PV PV PV PV PV PV PV PV PV	In-cabin operation (virtual)
[63]	2020	Gantry Crane	Wind speed Target load	Text Colored area (Orange)	NE E	PV PV	Remote operation (virtual)
[64]	2021	Mobile Crane	Lift capacity	Text	NE	SV	In-cabin operation (virtual)

(continued on next page)

Table 2 (continued)

Ref.	Year	Machine Type	Supplied Information	Corresponding Vis.	Vis. Embeddedness	Vis. Location	Test Scenario
[65]	2021	Mobile Crane*	Load weight	Text	NE	SV	In-cabin operation (in-situ)
			Maximum slewing radius	Text	NE	SV	
			Slewing radius	Text	NE	SV	
			Obstacles (Workers)	Colored area (Red)	E	SV	
			Obstacles (Workers)	Colored dot marker (Blue)	E	SV	
[66]	2021	Mobile Crane	Lift path (Planned)	Animation	E	PV	In-cabin operation (virtual)
			Slewing radius	Text	NE	SV	
			Lift capacity	Text	NE	SV	
			Load weight	Text	NE	SV	
			Clearance distance (to load)	Text	NE	SV	
			Clearance warning (to load)	Colored text highlight (Yellow/Red)	NE	SV	
[67]	2022	Mobile Crane	Maximum slewing radius	Colored arc (Red)	E	SV	In-cabin operation (in-situ)
[68]	2022	Tower Crane	Load fall zone	Colored area (Green)	E	SV	In-cabin operation (in-situ)
			Obstacles (Workers)	Colored area (Cyan)	E	SV	

* An offshore crane model in a manufacturing yard is classified as a mobile crane in this paper due to similar operational height and visibility constraints.

real-time flexibility are critical, given their frequent movement across various locations. In contrast, tower cranes require UIs that provide comprehensive situational awareness over large areas and extended operational periods, while gantry cranes demand UIs that emphasize precision in more predictable settings. Though each crane type presents specific challenges, these focal points are still essential considerations across all crane UIs, underscoring the need for thoughtful design that accommodates both common and unique operational needs. Section 3.3 goes on to highlight more universal design considerations for crane UIs.

3.2.2. Validation methods

Validation methods are critical for assessing the effectiveness and usability of crane UIs. This section explores the composition of user studies, participant types, and testing scenarios in the reviewed studies. It emphasizes the importance of involving real crane operators in user testing to gather reliable feedback and highlights how the testing environment may affect the applicability of the results. This discussion lays the groundwork for understanding how different validation strategies impact the overall reliability of crane UIs.

3.2.2.1. User studies and participant types. The selection of participants plays a critical role in ensuring that real-time UI systems for crane operations are validated with relevant, applicable feedback. Of the 18 studies reviewed, only 7 (39 %) directly involved professional crane operators in their evaluation processes. Recruiting operators for user studies proved challenging, often leading researchers to rely on smaller sample sizes. Most studies [33,50,51,53,56,62] included only 5–6 operators, with [65] engaging just 2 operators for validation. Despite these limitations, qualitative methods such as interviews and observations have still provided valuable insights into the practical considerations for UI design, including operator preferences and real-world usability challenges.

To supplement small operator samples, some studies included student participants to increase the pool size. For example, study [33] featured 30 students alongside 5 operators, and [25] exclusively relied on 30 students. While student feedback can offer insight from a novice perspective [69], the depth of feedback from students is often limited, as they lack the nuanced understanding of crane operation that seasoned professionals possess. This difference became clear in [33], where operators demonstrated a stronger focus on obstacle avoidance and efficiency during task execution, especially when using real-time assistance systems. In contrast, students tended to approach the tasks with less caution and lower operational awareness. Additionally, experienced operators showed greater reliance on communication with field workers when using conventional UIs, underscoring the importance of real-world experience in providing meaningful feedback.

Quantitative methods, such as NASA-TLX and the Technology Acceptance Model (TAM), used in studies like [33] and [50], respectively, have been employed to assess UI performance and usability. Additionally, study [54] distributed questionnaires over the course of three months, with results showing that 68.3 % of 162 users were satisfied with the system's performance, 24.4 % rated it as average, and 7.3 % provided negative feedback. Furthermore, 86.6 % of users believed the system could effectively prevent potential collisions, indicating strong perceived safety benefits. However, limited details are provided about the questionnaire itself, such as its structure or specific questions, leaving room for uncertainty about the depth of the feedback.

While small sample sizes and limited details on data collection methods pose challenges, the combination of qualitative and quantitative insights still offers valuable guidance for refining crane UI designs. These mixed methods remain crucial for addressing usability, safety, and effectiveness in real-time crane operations.

3.2.2.2. Test scenarios. The reviewed studies can be categorized into two primary operational contexts: in-cabin operation and remote operation. These are further divided into different testing environments: in-situ, laboratory, or virtual, where researchers tested their UIs under different levels of realism and control.

In-cabin operation refers to scenarios where the operator controls the crane from inside the cabin, closely mirroring typical real-world operation. Of the studies reviewed, 67 % (12/18) adopted this approach. Half of these studies (50 %) were conducted in-situ (e.g. [50,54,57,60,65,67]), where real-world factors like weather and complex terrain come into play, while the other half took place in virtual environments (e.g. [49,52,58,61,64,66]), providing a controlled setting to test core UI functionality without site-based complexities. Although virtual testing offers flexibility and efficiency, concerns about whether these simulations can replicate the intricacies of real-world operations persist.

In remote operation, the operator controls the crane from a distance, typically from a control room, relying on video feeds and sensor data. This category comprised 33 % (6/18) of the studies. Two studies [25,26] tested UIs in laboratories that offered controlled conditions that facilitated precise visualization placement, such as the printed markers used in [25] which are only effective at short ranges. However, this setup lacks real-world unpredictability. On the other hand, virtual environments (e.g. [51,63]), offer flexibility by simulating crane environments digitally, bypassing physical challenges but again raising questions about how well these results translate to actual operations.

Overall, the prevalence of virtual prototyping across 56 % (10/18) of the studies highlights the appeal of cost-effective, risk-free testing, with the ability to circumvent the technical complexities concerning AR

implementation in-situ [31]. Virtual environments enable rapid evaluations of UI designs and allow for innovative explorations, such as [63] visualizing first-person highlights for hidden loads. However, these benefits come at the expense of realism as virtual environments often do not implement lift scenarios at high levels of fidelity, which may limit the applicability of findings to real-world crane operations. In contrast, in-situ testing is invaluable for validating how UIs handle unpredictable site variables, though it faces technical hurdles such as aligning visualizations with physical elements [31].

Some studies (e.g. [51,63,67,68],) focused on testing individual visualizations rather than full UI systems. These visualization tests provide valuable insights into specific UI components but lack the comprehensive context needed for full system validation. Similarly, one study [62] used paper prototypes to gather early feedback on visualization placement. While useful, these simplified setups do not capture the cognitive demands of real crane operations, limiting their applicability for more advanced stages of validation.

While virtual and laboratory-based testing offers practical advantages in terms of cost, control, and rapid prototyping, the importance of in-situ testing should not be understated. Real-world environments offer unique challenges such as changing weather conditions, site layout, and other distractions that virtual testing has yet to fully replicate. A balanced approach that incorporates both virtual prototyping for early-stage design and in-situ testing for final validation may offer the best path forward for developing reliable crane UI systems.

3.2.3. Embedded and non-embedded visualizations

It is essential to first define the concepts of situated and embedded visualizations to provide context for the subsequent analysis. Situated visualizations are visual representations of data positioned near their physical referents, facilitating direct spatial associations for observers [70,71]. Embedded visualizations are a subset of situated visualizations which spatially integrate information more tightly with their referents, often using multiple physical presentations to display related data through methods like overlays, projections, and video see-through devices [71]. For example, both non-embedded and embedded visualizations might display the maximum slewing radius of a crane. In the former, this information may appear on a screen within the crane cabin as a numerical value. While this visualization is situated, it is not physically aligned with the corresponding referent, and thus its level of spatial integration is too low to be considered embedded. Here, observers must process and explicitly connect the visualization with the physical referent, splitting their visual attention between the two. In contrast, if the numerical value is projected around the crane's base, whether via a live camera feed or in the real world, its physical alignment with its referent would classify it as an embedded visualization as the visualization is able to be perceived simultaneously with the referent. These visualizations often employ other types of presentation modes such as graphics or symbols as they are easier to understand compared to text (illustrated in Fig. 3). This paper classifies embedded visualizations as not only those directly projected into the physical environment but also those closely integrated with digital representations of real-world objects. Although these visualizations may not align directly with physical referents, their alignment with digital counterparts that accurately represent real-world properties qualifies them as embedded.

Overall, the literature summary highlights a trend indicating that specific types of information are frequently conveyed through embedded visualizations, whereas others are more commonly depicted using non-embedded visualizations. Embedded visualizations emerge as the preferred choice for conveying critical information essential for immediate decision-making during crane operations. Clearance warnings, lift paths, load delivery zones, obstacles, and target loads are prominently featured in embedded visualizations across multiple research studies, highlighting their usefulness in enhancing situational awareness and facilitating rapid responses to changing conditions.

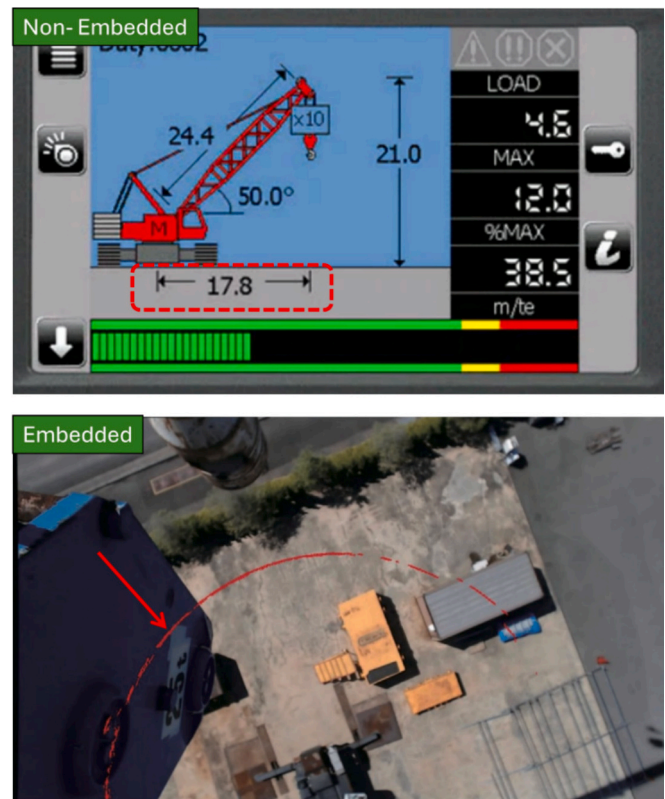


Fig. 3. Top: Non-embedded numerical visualization of operating radius on a typical Load Moment Indicator device – Modified from [42]; Bottom: Embedded graphical visualization of operating radius projected onto the real-world – Modified from [67].

Conversely, non-embedded visualizations are often employed for conveying supplementary information, such as textual displays of typical crane parameters such as boom length or load weight. While these details contribute to overall situational awareness, their importance typically becomes apparent only when nearing or surpassing critical thresholds, such as load capacity. This may be a reason as to why most studies opted for non-embedded modes of presentation for these types of information which are not immediately integrated into the operator's FOV.

This is especially evident in the qualitative findings of [33] which emphasize the prioritization of direct and critical information through embedded means such as AR overlays which helps operators to address immediate challenges. For instance, determining the next load movement direction or alignment, without the need for embedding extraneous supplementary information available through other views. However, the same study noticed an exception, where non-embedded text values of the current and target load heights were more useful to the operators compared to the 3D AR load path which may have been limited in precision. This suggests that operators may prefer to bear the cognitive cost of referring to non-embedded text values, potentially due to their higher perceived reliability and clarity, rather than the embedded visualizations. Consequently, improving the design of the load path visualization using improved depth cues (e.g., animation) or simply embedding the numerical values as part of an embedded visualization could offer a better solution which provides the most useful information while minimizing cognitive load. Nevertheless, the usefulness of any visualization is ultimately subjective in nature and depends on the ability of an individual to perceive a visualization and discern its meaning. This is noted by [62] who found that operators had different interpretations of the meanings of clearance warnings visualizations, and different preferences for crane balance visualizations. Although

experience or training with a system can mitigate this issue over time, it emphasizes the need for visualizations to be simple and intuitive to ensure consistent messaging.

Apart from effectiveness in real-time decision-making and operator preferences, the decision to embed visualizations to use may also depend on technological feasibility. Implementing these visualizations in-situ typically requires specialized AR equipment or complex Internet-of-Things (IoT) sensor networks to create a dynamic digital representation of the lift environment as noted by [60]. The relative technological immaturity of the hardware and software infrastructure necessary for

embedded visualizations likely lessens their prevalence compared to non-embedded alternatives. Overall, the distribution of embedded and non-embedded information in the literature reflects a complex interplay between their effectiveness in conveying critical information, operator preferences, and technological capabilities in the context of crane operation.

3.2.4. Visualization placements

In addition to the type of visualization, its placement is also a pivotal consideration, impacting an operator's situational awareness and

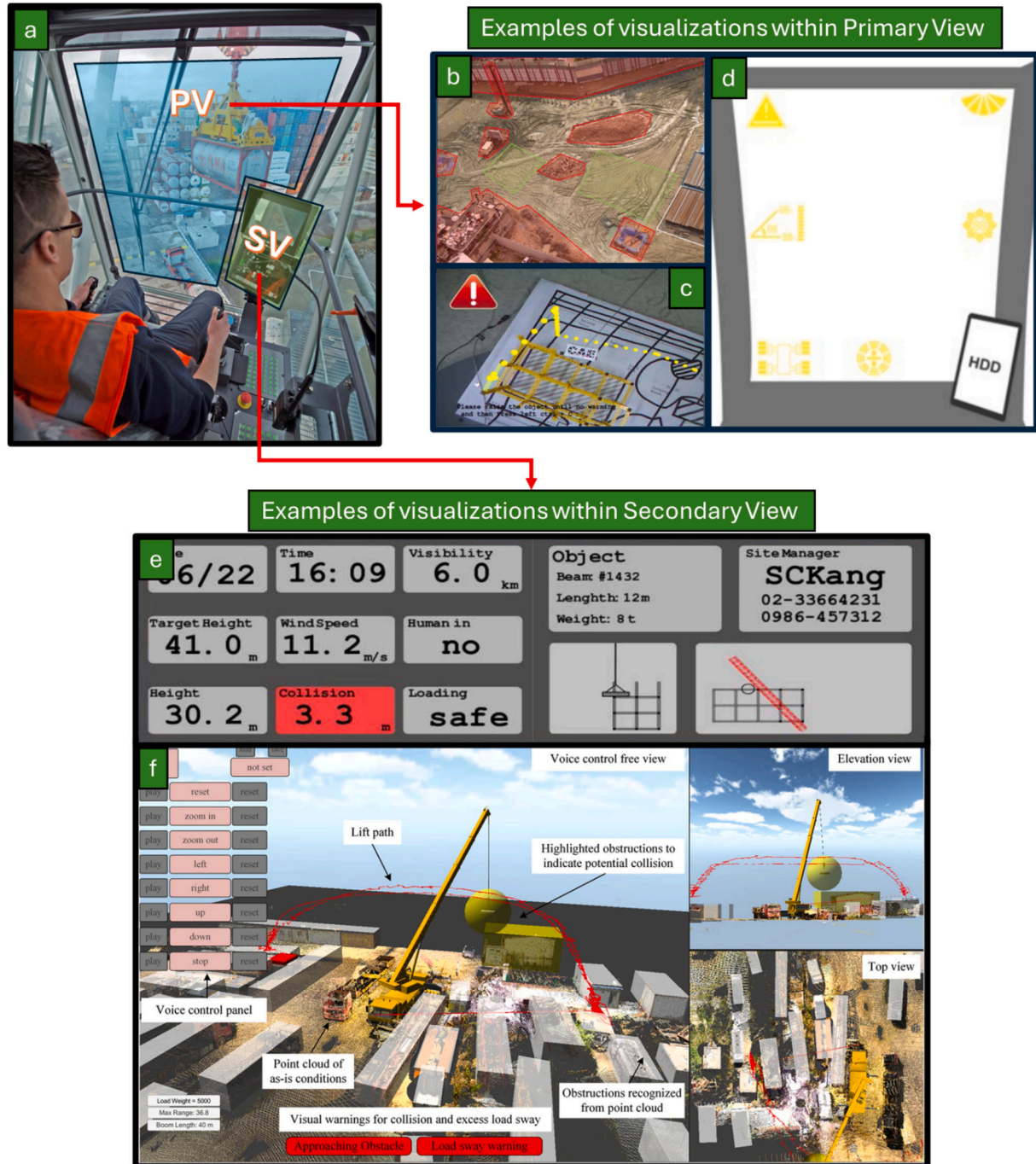


Fig. 4. Examples of visualizations within Primary and Secondary Views: a – Estimated areas representing the Primary View and Secondary View within a crane cabin – Modified from [76]; b – Embedded visualizations of obstacles and safe working area [49]; c – Embedded visualizations of recommended load path and load delivery zone, Non-embedded visualization of alert symbol [25]; d – Non-embedded visualizations of various crane parameters on a windshield display [62]; e – Non-embedded visualizations of various crane parameters on a supplementary information display [25]; f – A mix of non-embedded and embedded visualizations overlaid on a digital representation of a site [57].

consequent performance. Studies like [72] have suggested that positioning visualizations closer to the center of an operator's FOV increases their likelihood of being perceived, reducing the need for operators to divert attention to peripheral displays. This makes sense when considering how human vision functions: only the small central region of the visual field, the fovea, allows for high-resolution detail perception [73]. This area is densely packed with light-sensitive receptors that process detailed information, whereas the periphery of our vision captures much less detail. As a result, placing visualizations in the primary visual area (i.e., within an operator's central FOV) optimizes the likelihood of rapid detection and clear interpretation, minimizing the need to shift focus away from crucial tasks.

As such, this paper classifies visualization placements in the reviewed literature into two main categories: "Primary View" (PV) and "Secondary View" (SV). PV visualizations are located within the crane operator's typical FOV, centered around the hook and load where their attention is naturally focused during a lift. In contrast, SV visualizations are presented on separate displays, accessible but not immediately within their typical line of sight. Fig. 4 illustrates a typical representation of PV and SV areas within a crane cabin and provides some examples of visualizations that studies have explored. This research applies this classification equally to studies conducted in digital simulations (e.g., Virtual Reality or VR) and the physical world, as it can be argued that an operator's perception of visualizations remains effectively the same in both contexts. This is because VR technology can accurately replicate the spatial and visual conditions of real-world environments effectively [74]. Moreover, VR allows for controlled testing environments and consistent repetition of scenarios [75], ensuring that the findings on visualization effectiveness are robust and generalizable to real-world operations.

3.2.4.1. Primary view (PV). The placement of PV visualizations within the operator's typical FOV increases their likelihood of being perceived, as the operator's vision naturally passes over them more frequently [72]. The reduced time spent searching for information allows operators to maintain focus on load handling and their surroundings. This implies that PV visualizations should present critical, time-sensitive information critical for immediate decision-making, similar to trends noted for embedded visualizations [33].

One approach to placing PV visualizations is through transparent windshield displays, as demonstrated by [61], where LED arrays depict the crane's boom length, operating radius, and capacity level. However, these displays do not necessarily embed the visualizations with their physical referents, requiring operators to expend additional cognitive effort to associate the data and their real-world counterparts [57]. While this effort may be manageable with simple data, it can become problematic with more complex information. For example, crane operator interviews conducted by [57] revealed that although operators recognized the potential benefits of PV visualizations, they also encountered instances where the visualizations did not accurately represent the referent objects. Specifically, non-embedded visualizations for clearance warnings often failed to provide a clear spatial picture in situations with multiple obstacles.

In addition to the complexity of the data, operator interviews in [62] emphasized the importance of keeping the center of the PV clear of visualizations to avoid obstructing the load, which is crucial for safe crane operations. These operators generally preferred positioning visualizations within the peripheral areas of the PV presumably to avoid visual clutter. However, this preference likely stems from the fact that the non-embedded visualizations used were more intrusive and harder to interpret intuitively, leading operators to favor peripheral placement to minimize disruption. In contrast, embedded visualizations which seamlessly integrate with physical referents, offer a more intuitive and less obtrusive way to present critical information. As a result, these visualizations can be placed more centrally without causing the same level

of distraction or visual clutter. This suggests that with well-designed embedded visualizations, operators might be more open to accepting central PV placements, as the visual information would enhance rather than hinder their focus on the load and surrounding environment.

Nevertheless, developing transparent displays that are able to embed visualizations in the PV remains a significant challenge in real-world crane operation settings, as evidenced by the scarcity of studies addressing this issue. Commercially available Head-Mounted Displays (HMD) such as the Apple Vision Pro and Microsoft HoloLens are devices that have seen successful applications in close-range applications such as surgical operations [77] and construction inspection works [78]. However, it is clear that these devices are not a preferred option among current crane operation research due to several implementation issues [32]. Specifically, there are major concerns pertaining to registration errors at long ranges which will lead to inaccurate visualization positioning. Additionally, the limited FOV of these devices restricts their effectiveness as operators are not able to utilize their peripheral vision effectively. Smaller issues with the weight of the device and battery life, are also exacerbated noting an operator's long working hours.

An alternative method utilizes Spatial Augmented Reality (SAR) [79], an emerging technology that projects visualizations directly onto physical referents. SAR has the potential to create effective embedded visualizations visible to both the crane operator and secondary lifting personnel, such as signalmen and riggers. However, its application in real-world construction environments is also limited by technological challenges, such as maintaining continuous alignment between the visualizations and their physical referents. As a result, researchers have instead explored these visualization placements in smaller-scale remote operation studies (e.g. [25,26],) and VR environments (e.g. [64,66],), where the technological demands are less stringent. For example, visualizations such as lift paths and object highlights can be embedded onto their referent objects within a digital space such as a game engine environment like in [66] to achieve a similar effect to SAR projections onto an object in the real world. In the same vein, the recreation of transparent displays or other types of HUD is also more achievable in these controlled environments as less specialized hardware is required.

3.2.4.2. Secondary view (SV). In contrast to PV visualizations, SV visualizations require operators to shift their attention away from their primary task, potentially increasing cognitive load [72]. Traditionally, these visualizations present supplementary information, such as crane parameters (e.g., slewing angle, boom angle, load weight, load height), on LMI displays. Because this data is often numerical and non-embedded, it can hinder operators' ability to make timely and accurate decisions. The peripheral placement of these displays also means they are less likely to be noticed unless operators deliberately seek them out. However, as noted by [58], the effectiveness of SV displays may depend more on the relevance and clarity of the content presented rather than their exact physical placement within the peripheral area. This suggests that well-designed, easily interpretable content could mitigate some of the cognitive challenges posed by their location.

In modern cranes, it has become commonplace to have an additional top-down camera view located above the hook to help improve an operator's depth perception and situational awareness. These views are typically made available next to or built into SV displays such as the traditional LMI. Notably, these additional views allow the operator to continue maneuvering the load despite not looking at it directly. Some studies have expanded this approach by providing camera views from other angles [25,33]. While these views offer valuable perspectives, their practical implementation in real-world scenarios is challenging due to the need for frequent repositioning of external cameras as construction progresses. Instead, studies have explored using dynamic digital representations of the site environment which effectively function as alternate views to help improve situational awareness.

Digital representations range in complexity from simple annotated

2D images of crane models [50] to full 3D site models [57] created using reality capture technology. One study [50] found that operators generally preferred 2D digital representations of crane orientation over 3D representations, as the 2D format made it easier to discern the precise location of the load. While the study did not delve into the reasons behind this preference, an analysis of the provided UIs suggests that this preference is due to the embedding of crane parameter data within the 2D representations, which was not the case with the 3D representations. Conversely, other studies [56,57,65] introduced SV displays utilizing 3D digital representations with embedded visualizations, which operators found particularly useful for lifts where visibility is a concern, such as blind or night lifts. Although no direct comparison was made with 2D representations in this study, the positive feedback indicates that well-designed 3D representations with embedded visualizations can effectively support complex lifting tasks.

Digital representations are prevalent for several reasons. For instance, digital representations can be easily updated to reflect changes in the site environment as a project progresses with the availability of accurate BIM data in this era [80]. Moreover, these representations offer enhanced customization options, such as adjustable view angles and zoom levels, allowing operators to tailor their viewing experience to the specific demands of each lift task – something that is difficult to achieve with traditional video feeds alone due to the rigidity of typical camera installations. Additionally, it is generally easier to implement and align visualizations precisely with their referents within a purely digital environment, as the exact spatial coordinates and dimensions are known and can be manipulated directly. However, when digital models are based on real-world sensor data, the process becomes more complex. Issues like sensor calibration, data latency, and environmental conditions can introduce challenges in maintaining accurate alignment, particularly in dynamic conditions [31].

Interestingly, some research [57,65] has demonstrated that displays with highly detailed digital representations can enable operators to conduct crane operations without a direct view of the physical environment. In these scenarios, the SV effectively becomes a PV in terms of the operator's focus. This was evident in their experiments where operators operated the crane solely through an SV display, with their windshield obscured. The success of such experiments underscores the potential for SV displays to be designed in a way that fully immerses the operator, effectively converting what is traditionally considered secondary information into a primary source for decision-making.

Furthermore, it was also noted that none of the reviewed studies conducted in virtual environments attempted to create separate digital representations of the site environment before overlaying visualizations. Instead, they projected visual information directly onto the digital scene, simplifying the process. This approach makes sense, as recreating a digital version of an already virtual environment would add unnecessary complexity. Whilst operators may appreciate the direct overlays of visualizations onto the virtual environment, such visualizations are yet achievable in the real world. This suggests that researchers need to consider whether direct overlays, while easier and seemingly more practical, are useful in this stage when the technology to recreate these visualizations are still in their infancy.

Despite the promise of these advancements, several issues have emerged. For instance, some studies [57] have observed that operators may become distracted by SV displays, leading to failures in managing critical tasks like load sway. This distraction could arise from unclear visual cues within the digital representations, which may not make certain hazards immediately apparent. Consequently, operators might struggle to anticipate or correct these issues in a timely manner, even when they spend additional time on maneuvers. The same study also highlighted trust issues with the interface, particularly when operators' direct view of the load through the windshield was obscured. Under these conditions, their average lift times doubled compared to regular operations as they were more cautious. Although lift times may improve with extended use as operators gain familiarity, it is yet clear as to what

extent this trust issue will remain a factor. Further complicating matters [65], emphasized that the effectiveness of additional views depends on the specific task's context and the support required. For instance, a typically provided top-down view, was deemed "not helpful at all" by two operators during a blind lift, as it failed to assist with visualizing load height – the most uncertain parameter in the operation. Although a more useful side view was provided for this purpose, its implementation introduced confusion when the load spun, causing the entire view to rotate with it.

Overall, current research still needs refinement to determine the best way to use PV or SV displays for optimal performance. The studies reviewed cover a wide range of topics, and since this area of research is still developing, it is too early to draw definitive conclusions about the ideal placement of visualizations. Rather, the current positioning of visualizations is likely influenced more by the ease of implementation and the technical constraints faced by researchers, rather than by a clear understanding of where they would be most effective. The key to effective visualization placement is ensuring that operators can easily access critical information without losing focus on their primary task of load handling. Thus, it seems that the main priority is not whether the information is placed in the PV or SV but ensuring that whichever display is used provides all the necessary information to maintain the operator's spatial awareness and support safe operations.

3.3. UI evaluation

Table 3 presents the application of the evaluation model to critically evaluate the UIs featured in each of the research groups (if available). In total, 14 UIs are evaluated using the model. This comprehensive evaluation enables a thorough examination of the state-of-the-art literature, shedding light on the strengths, weaknesses, and areas for improvement in real-time assistance systems for crane operators. By systematically assessing the quality of recent studies, identifying gaps in knowledge, and reflecting on the validity and reliability of research findings, this analysis results assist in understanding the landscape of crane operation systems. Additionally, two spider charts illustrate the observed percentage of the design principles for each principle group and for each UI (Fig. 5). These visualizations provide a clear and comparative overview of how well each study adheres to the established design principles, highlighting areas where further improvements are needed.

3.3.1. Consistency

3.3.1.1. Leveraging occupational familiarity. Occupational familiarity was observed in 78.57 % of the reviewed UIs. This principle involves aligning the interface with established conventions and operational habits, enhancing usability by building on operators' pre-existing knowledge. In UIs where this principle was observed, such as in [56,57], the UI effectively functioned as a replacement to the traditional LMI, incorporating the typical information expected on the device. This allowed operators to quickly adapt to its use with a minimal learning curve. Conversely, unobserved instances of this principle were found in UIs that introduced new interaction methods or unfamiliar elements. For example [49,55], employed different colored highlights to emphasize areas of interest, but the unconventional use of white and blue lacked an intuitive connection to established color conventions, where red and yellow typically indicate danger or caution. This deviation could lead to confusion or hesitation, as operators may struggle to interpret the meaning of the colors. Similarly [26], featured a simple prototype interface designed as a collision detector, but its setup required additional steps, potentially disrupting the operator's workflow.

3.3.1.2. Stability of components. Stability of components was observed in all of the UIs reviewed. This principle requires maintaining the consistent placement and function of critical UI elements, allowing

Table 3
Evaluation Model Result Summary.

Principle Category	No.	Principle	[49]	[26]	[50]	[52]	[25,33]	[53,54]	[55]	[56,57]	[58]	[59,60]	[61,62]	[64]	[65]	[66]	Observed
Consistency	1	Leveraging Occupational Familiarity: •Incorporate familiar crane operation symbols and conventions.			✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	78.6 %
	2	Stability of Components: •Ensure consistent positioning and functionality of UI components.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100.0 %
	3	Uniform Design for Similar Components •Ensure visual consistency among similar UI components.	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	92.9 %
	4	Aesthetic and Readable Designs •Create visually pleasing designs. •Ensure readability in varied conditions.			✓		✓	✓		✓	✓	✓	✓	✓	✓	✓	71.4 %
	5	Context-Aware Designs •Prioritize critical information based on operational context. •Minimize clutter.		✓			✓	✓	✓	✓			✓		✓		50.0 %
Clarity	6	Distinct and Intuitive Visual Elements •Use clear graphics for easy interpretation.			✓		✓			✓	✓	✓	✓		✓	✓	57.1 %
	7	Multiple Views for Complex Elements •Provide alternative perspectives during lifts.		✓	✓	✓	✓	✓		✓	✓	✓		✓	✓		71.4 %
	8	Use of Color and Graphics as Perception Aids •Leverage color and graphics to highlight critical information.	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓		✓	✓	85.7 %
Organization	9	Virtual Object Integration •Integrate virtual elements seamlessly. •Prevent obstruction of critical information.		N/A	N/A		✓	N/A		✓	✓	✓		N/A	✓	✓	60.0 %
	10	Structured Information Delivery •Sequence information delivery logically. •Signal clear transitions for operators.					✓	✓		✓		✓				✓	35.7 %
	11	Logical Grouping of Information and Controls •Arrange information and controls hierarchically. •Establish clear relationships between elements.			✓		✓	✓		✓	✓	✓	✓	✓	✓	✓	71.4 %
	12	Manage User Attention •Direct user attention to critical elements.		✓			✓	✓	✓	✓	✓	✓	✓		✓	✓	71.4 %
	13	Automation of Workload •Streamline mental calculations and estimations.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100.0 %
Usability and Compatibility	14	Flexible Data Presentation •Allow configurable interface settings. •Cater to different user preferences.			✓							✓			✓		21.4 %
	15	Simplicity of Controls •Minimize number of controls. •Minimize eye and hand movements.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100.0 %

(continued on next page)

Table 3 (continued)

Principle Category	No.	Principle	[49]	[26]	[50]	[52]	[25,33]	[53,54]	[55]	[56,57]	[58]	[59,60]	[61,62]	[64]	[65]	[66]	Observed
Error Prevention and Recovery	16	Feeling of Control • Instill a sense of user control. • Ensure predictability and responsiveness of UI.				✓											7.1 %
	17	Collaborative Interface • Allow for effective communication and coordination between lift personnel. Protection Against Serious Errors															0.0 %
	18	• Implement preventive measures to safeguard against critical errors.		✓			✓	✓	✓				✓			✓	64.3 %
	19	Error Resolution • Provide clear error messages with actionable feedback.				✓						✓					14.3 %
Observed			26.3 %	50.0 %	61.1 %	31.6 %	89.5 %	72.2 %	36.8 %	78.9 %	63.2 %	84.2 %	63.2 %	50.0 %	73.7 %	68.4 %	

operators to quickly locate and interact with key information during crane operations. Given that most crane interfaces are designed with a focus on simplicity and predictability, incorporating stable components becomes an inherent feature. In fact, developing a UI where critical elements do not maintain consistent positions and functions would likely require significant additional effort. Thus, the consistency observed in the analyzed UIs reflects an efficient approach to ensuring usability and reducing cognitive load for crane operators.

3.3.1.3. Uniform design for similar components. Uniform design for similar components was observed in 92.86 % of the UIs. This principle emphasizes avoiding casual differences in the visual design of components with similar functions (e.g., fonts, colors), ensuring that operators can easily identify and interact with elements related to specific crane functionalities. The only instance where this principle was unobserved was in [55], where red was used to highlight a clearance issue on the UI while blue was used to indicate the obstacles with the clearance concerns. This lack of consistent visual messaging might impair the operator's ability to quickly and accurately interpret whether a hazard is occurring or not. Overall, the adherence to uniform design principles helps streamline interactions and supports effective operational decision-making.

3.3.2. Clarity

3.3.2.1. Aesthetic and readable designs. Aesthetic and readable designs were observed in 71.43 % of the UIs. This principle highlights the need for clear and visually appealing interfaces that enhance readability and minimize cognitive strain. UIs that effectively observed this principle featured appropriately sized text and high color contrast, ensuring that information was clearly legible. In contrast, UIs such as those in [26,52,55] did not meet this principle's criteria, as they included visual elements with text that was too small or had poor color contrast, impairing readability. Ensuring aesthetic and readable designs is essential for operators to quickly and efficiently process information, especially in the demanding environment of crane operations.

3.3.2.2. Context-aware designs. Context-aware designs were observed in just 50 % of the reviewed UIs. This principle involves adapting the UI to prioritize critical information based on the current operational context as research has shown that operators are more likely to perceive visualizations that directly support the task they are actively engaged in [72,73]. UIs such as those in [25,33,56,57] effectively observed this principle by making important data more prominent during high-risk scenarios. In contrast, UIs like those in [58–60] presented static designs with constant information visibility, which could lead to information overload and make it challenging for operators to filter out essential data during critical moments.

3.3.2.3. Distinct and intuitive visual elements. Distinct and intuitive visual elements were observed in 57.14 % of the reviewed UIs. This principle focuses on using simple and easily understandable symbology to aid quick interpretation of information. UIs such as [61,62] employed simple, intuitive symbology that allowed operators to easily interpret the information presented at first glance. However, UIs in [49,55] did not observe this principle effectively, as they used colored highlights without providing a sufficient explanation of what each color represents, potentially increasing cognitive load and complicating quick interpretation.

3.3.2.4. Multiple views for complex elements. Multiple views for complex elements were observed in 71.43 % of the reviewed UIs. This principle supports the use of additional views to aid spatial awareness and understanding of complex crane tasks by supporting an operator's depth perception of the load. Many UIs provide these views in the form of live

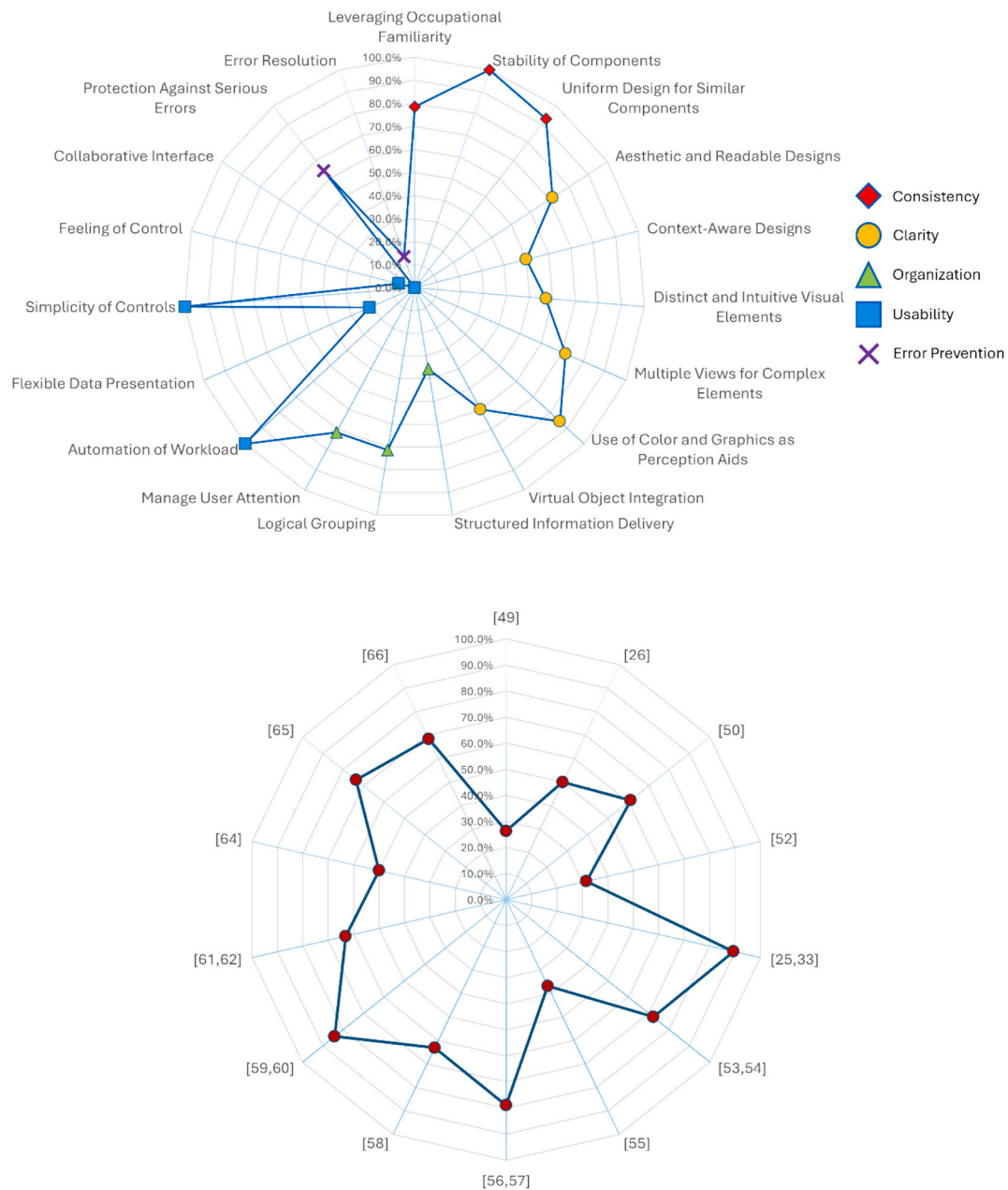


Fig. 5. Summary of observed % of design principles (top – within each principle group, bottom – within each UI).

camera or video feeds [51,52], digital representation equivalents [26], or both [25,33,53,54] to offer diverse perspectives. These designs were effective in helping operators gain a comprehensive understanding of the operational environment around the load. To a lesser extent, studies [49,63] explored alternative ways to facilitate depth perception by highlighting objects hidden behind obstacles with visualizations. However, it is not clear whether these methods are as effective as the provision of additional views.

3.3.2.5. Use of color and graphics as perception aids. This principle was observed in 85.71 % of the reviewed UIs, underscoring the importance of leveraging color and intuitive graphics to guide the operator's attention effectively. Most UIs made effective use of high contrast colors to highlight critical areas and alert the operator to potential hazards. For example, colors like yellow or red were used to indicate clearance issues, or proximity to obstacles, making it easier for operators to notice them quickly without the need for extensive cognitive effort. Conversely, UIs

such as [52,64] did not observe this principle effectively. In these studies, there was little effective use of color to direct attention and an overreliance primarily on text to communicate crane parameters. The absence of color-coded alerts or visual cues likely increased the operator's cognitive load, requiring them to spend more time scanning and interpreting the information manually. This approach reduces the UI's effectiveness in high-pressure construction environments, where quick decision-making is critical.

3.3.3. Organization

3.3.3.1. Virtual object integration. Virtual Object Integration was observed in 60 % of the 10 UIs that included digital overlays over the site environment. This principle emphasizes the mindful placement of digital information such that it is clear to the operator but does not obscure critical elements of the environment. UIs in studies such as [25,33,59,60] followed this principle effectively by positioning virtual

components in ways that allowed operators to maintain visibility of crucial operational surroundings. Conversely, UIs in [49,52,55] failed to clarify whether virtual objects would obstruct important elements in the environment, potentially compromising situational awareness and leading to operational errors. Notably [61], considered transparent displays where the opacity of the visualizations may be adjusted, but it is unclear whether this method is effective due to the lack of strong validation.

3.3.3.2. Structured information delivery. This principle was observed in just 35.71 % of the reviewed UIs. Structured information delivery refers to the sequencing of feedback in a way that mirrors the stages of an operation, such as the start, progress, and completion of a crane maneuver. UIs like those in [25,33,53,54] employed this principle to some extent, with features like color-coded indicators, or timed notifications that kept operators informed about the status of a lift in real-time. This structured feedback helps prevent confusion, allowing operators to better anticipate the next steps in the operation. On the other hand, the majority of the UIs did not integrate such structured information delivery, resulting in a more static presentation of data. Without clearly delineated feedback for different stages of crane maneuvers such as a progress bar, operators will have to rely more on their own experience or intuition.

3.3.3.3. Logical grouping of information and controls. Observed in 71.43 % of the reviewed UIs, logical grouping organizes related information together, facilitating quick access and interpretation for operators. While this is observed in many studies, some failed to observe this principle effectively. For example [26], did not account for the presentation of critical crane parameters clearly in their design, which may have led to confusion and slower response times. Prioritizing logical grouping is crucial for reducing cognitive load and improving operational efficiency in crane interfaces.

3.3.4. Usability and compatibility

3.3.4.1. Manage user attention. This principle was observed in 71.43 % of the reviewed UIs and focuses on guiding operators toward critical information using various cues (visual, auditory, or haptic). Effective implementations, such as in [26,55], utilized strategies like color changes and audio alerts to draw attention during significant operational events such as hazards, ensuring that operators are alerted to their presence. However, 28.57 % of the UIs failed to incorporate effective attention management techniques, with studies [49,50,52,64] highlighting a lack of cues that left operators reliant on their vigilance. This absence of guidance can result in missed warnings or critical updates, increasing the risk of accidents during crane operations.

3.3.4.2. Automation of workload. This principle was observed in 100 % of the reviewed UIs, highlighting that their primary purpose is to streamline information delivery and minimize cognitive load for operators. Effective UIs automatically estimate critical parameters, such as load height, fall zones, and clearances, allowing operators to focus on operational tasks without needing to calculate these factors manually.

3.3.4.3. Flexible data presentation. This principle was observed in only 21.43 % of the reviewed UIs, indicating a limited emphasis on adaptability in information display. Effective UIs, such as those in studies [50,59,60,65] provided configuration options that operators could modify during operation, allowing for a more personalized experience. However, many interfaces lacked this flexibility, which may restrict operators' ability to tailor the information presented to their specific needs, potentially hindering decision-making efficiency.

3.3.4.4. Simplicity of controls. This principle was consistently observed

in all reviewed UIs (100 %). Emphasizing simplicity in controls is vital for effective crane operations as operators typically need to keep their hands on the joystick controls. Interfaces typically feature minimalistic designs, where interaction with a system is kept low during maneuvers barring some utilizing simple touch screen interfaces or voice activation [56,57]. By reducing the number of controls, these UIs enhance usability, allowing operators to focus on critical tasks without being overwhelmed by complex input methods.

3.3.4.5. Feeling of control. This principle was only observed in 7.14 % of the reviewed UIs. Most studies do not sufficiently address usability aspects related to this principle, making it challenging to assess operators' feelings of control during crane operations. The limited coverage suggests a gap in the evaluation frameworks, indicating that the sense of agency and confidence in system interactions is often overlooked. A more focused investigation into this aspect could enhance UI design by ensuring operators feel empowered and in command while using the systems.

3.3.4.6. Collaborative interface. This principle was not observed in any of the reviewed UIs. The absence of collaborative properties in crane interfaces indicates a significant missed opportunity to enhance teamwork. Seamless information sharing among lift personnel is essential for safety and operational efficiency. The integration of collaborative features could foster better coordination and communication during crane operations as the lift crew will be able to develop a shared mental model of the lift, ultimately leading to improved situational awareness and decision-making [63].

3.3.5. Error tolerance and recovery

3.3.5.1. Protection against serious errors. This principle was observed in 64.29 % of the reviewed UIs. Effective measures to signal against or prevent dangerous crane operations were implemented in various interfaces, such as those in [25,33,56,57]. These UIs included features like alerts and warnings that help operators recognize potential hazards and avoid serious errors during operations. However, the presence of such protective measures was not universal, suggesting that many UIs still lack essential safety features. This inconsistency highlights the need for greater emphasis on error prevention in crane interface design to ensure operator safety and operational integrity.

3.3.5.2. Error resolution. This principle was observed in 14.29 % of the reviewed UIs. Effective error resolution features were present in some interfaces, such as those in [25,33,59,60], which provided clear error messages and guidance for resolving issues. These mechanisms are crucial for helping operators understand problems quickly and take corrective actions, thereby minimizing downtime and enhancing operational efficiency. However, the low prevalence of this principle indicates a significant gap in many UIs, where operators may not receive adequate support during error situations. Addressing this shortcoming could improve overall usability and safety in crane operations.

4. Decision framework and future research directions

This section aims to provide a detailed overview of recommendations for future research. The first part of this section introduces a detailed decision-making framework designed to guide researchers in the systematic development and evaluation of real-time crane operation UIs. Researchers are encouraged to build upon or adapt this framework to suit their specific needs. The second part of this section highlights specific gaps in current research that demand more specialized attention or are potentially highly lucrative, aiming to address unresolved issues, explore new technologies, and improve the overall effectiveness and safety of crane operations based on the findings from the earlier review.

4.1. Decision framework for real-time crane lift assistant user interfaces

The development of real-time crane lift assistant UIs is a complex process that requires careful consideration of various operational, design, experimental, and evaluation factors. To address this complexity, this paper proposes a comprehensive decision-making framework (Fig. 6) to guide future researchers in systematically developing and assessing UIs for real-time crane operations.

The first stage, Operational Specifications, outlines fundamental operational parameters that must be considered before initiating UI design. It differentiates between in-cabin and remote operations, as well as the various crane types, including tower cranes, mobile cranes, and gantry/overhead cranes. Establishing these specifications early on is a requirement noting the unique implementation and operational challenges associated with each operation type and crane type.

The second stage, UI Design, focuses on the design aspects of the user interface itself. In this stage, researchers are highly encouraged to utilize the evaluation model presented earlier in this paper (Table 1) to inform their design choices based on well-established design principles tailored to a crane operation context. This stage largely comprises of developing the individual designs of each visualization to be supplied to operators. Researchers must consider a few factors, starting with what information they wish to supply. This paper recommends building on a list of minimum essential information types including what is typically available to them in traditional LMI systems. This includes details of crane capacity, load mass and height, boom angle and length, as well as the current slewing radius which will ensure operators have the same critical data needed to make informed decisions akin to their daily operations. Next, it is important to decide where this information should be displayed to the operator, whether through a PV (e.g., see-through displays) or SV (e.g., video feeds or digital representations) display. Additionally, the degree to which the information is embedded within the operator's view must be considered noting that embedded information can potentially enhance situational awareness and reduce cognitive load. This should be considered together with the corresponding visualization methods such as text, graphics, overlays, or animations with a focus on intuitive designs that operators can easily interpret and act upon.

Additionally, the UI Design stage also considers what operation perspectives will be made available to the operator, with additional views to facilitate depth perception such as a typical top-down hook camera view or other angled perspectives. The final factor considered in this stage is the interaction method operators will use to work with the UI. Researchers must consider that operators delegate most of their attention to manipulating crane and load movements during operations; therefore, interactions with the interface should be kept to a minimum or made very convenient to avoid distractions. Possible methods include voice commands, simple touch screens, or other push-button interfaces located near or on the control joysticks.

In the third stage, Experimental Setting, researchers need to carefully consider their choice of testing environment and participants for validation purposes. Researchers have the option of using virtual environments or physical environments. Virtual environments are often employed to circumvent the risks associated with actual crane operations and technological hurdles of physical prototypes, allowing for quick modification of parameters and shorter design iteration periods. Similarly, laboratory model operations minimize risks while providing a controlled environment where physical prototypes of the assistant UI can be tested albeit at a smaller scale. Full-scale in-situ operations, on the other hand, offer real-world validation but can reveal technological, social, and economic challenges that may not be apparent in a virtual setting. Each environment has its merits, and a combined approach may offer the most robust evaluation.

Besides that, this stage also takes into account the selection of participants, which is critical for gathering diverse and relevant feedback. Involving professional crane operators is essential to verify the

behavioral responses of the end users and allows researchers to gain insights into design flaws based on the tacit knowledge of the operators. Typically, past research has involved 5–6 experienced crane operators, citing recruitment issues. While this number of participants may be able to provide valuable qualitative responses, a larger sample size may be needed to capture variability in operator preferences and behaviors. This is particularly important for understanding how different operators might prefer visualizations to be presented. A good number for such studies could range from 20 to 30 participants which would be more functionally representative of the target user group and also allow for more reliable statistical analyses of quantitative data to be conducted. Another option involves including student operators or novices to supplement participant numbers. Novice perspectives can be particularly useful in scenarios where operating experience is not a critical factor. For example, tasks that focus on the accessibility and usability of the UI, such as interpreting visualizations, and using interaction methods (e.g., touch screens or voice commands), do not necessarily require extensive crane operation experience. Novices can provide valuable feedback on how easily they can learn and work with the UI, which is crucial for designing user-friendly systems.

The last stage, Evaluation and Validation, focuses on the types of evaluation metrics that can be used to assess operator behavior and the supplementary methods that can be used to validate results depending on the study's objectives. Researchers may use metrics like cognitive load, which can be quantified using the NASA-TLX which measures perceived workload across several dimensions or heart rate. Gaze behavior, another metric that has surprisingly seen very limited investigation in UI design overall [25,58], involves analyzing the duration and frequency of gaze fixations on specific UI elements. This has high potential as a non-biased subjective indicator in evaluating their intuitiveness and effectiveness [81]. By analyzing where and how long operators focus their attention, researchers can gain valuable insights into which elements of the UI are most engaging or confusing, helping to identify areas for improvement. Other metrics such as task performance should be evaluated through efficiency and safety factors, including the time taken to complete tasks, accuracy of operations, and the number of errors or safety incidents. These metrics provide objective data on the UI's support for efficient and safe operations.

This stage also considers the methods used to validate results. Cross validation methods such as questionnaires and interviews to gather operator opinions are commonplace across many studies. Researchers should take this opportunity to gather deeper insights with open-ended questions that focus on an operator's experiences during the experiment to note possible improvements to the UI or even the experimental environment. Notably, measuring an operator's feeling of control is an often-overlooked aspect that can significantly impact the adoption rate of a system in the future. Apart from individual validation methods, focus groups are also an option to consider when handling larger sample sizes to condense the ideas of many participants and reach a consensus. Alternatively, longitudinal analysis is another valuable method, as it involves observing participants over an extended period. This approach helps to identify performance benefits that may not be immediately evident as participants gain familiarity with new systems, providing a better idea of the long-term effectiveness of UIs. Comparative studies can also be employed to evaluate different UI designs against each other, offering a clearer understanding of which elements are most effective in supporting efficient and safe crane operations. By using a combination of these methods, researchers can ensure a thorough and robust validation of their findings.

By covering the fundamental elements, this framework serves as a foundation for future studies to build upon, adapt, and expand. The examples given for the components in each stage are suggestions based on past studies which can be tailored to fit various contexts and are by no means an exhaustive list. Supplementary considerations will likely be necessary for specific research objectives which investigate new information types, displays, interaction methods, or evaluation metrics to

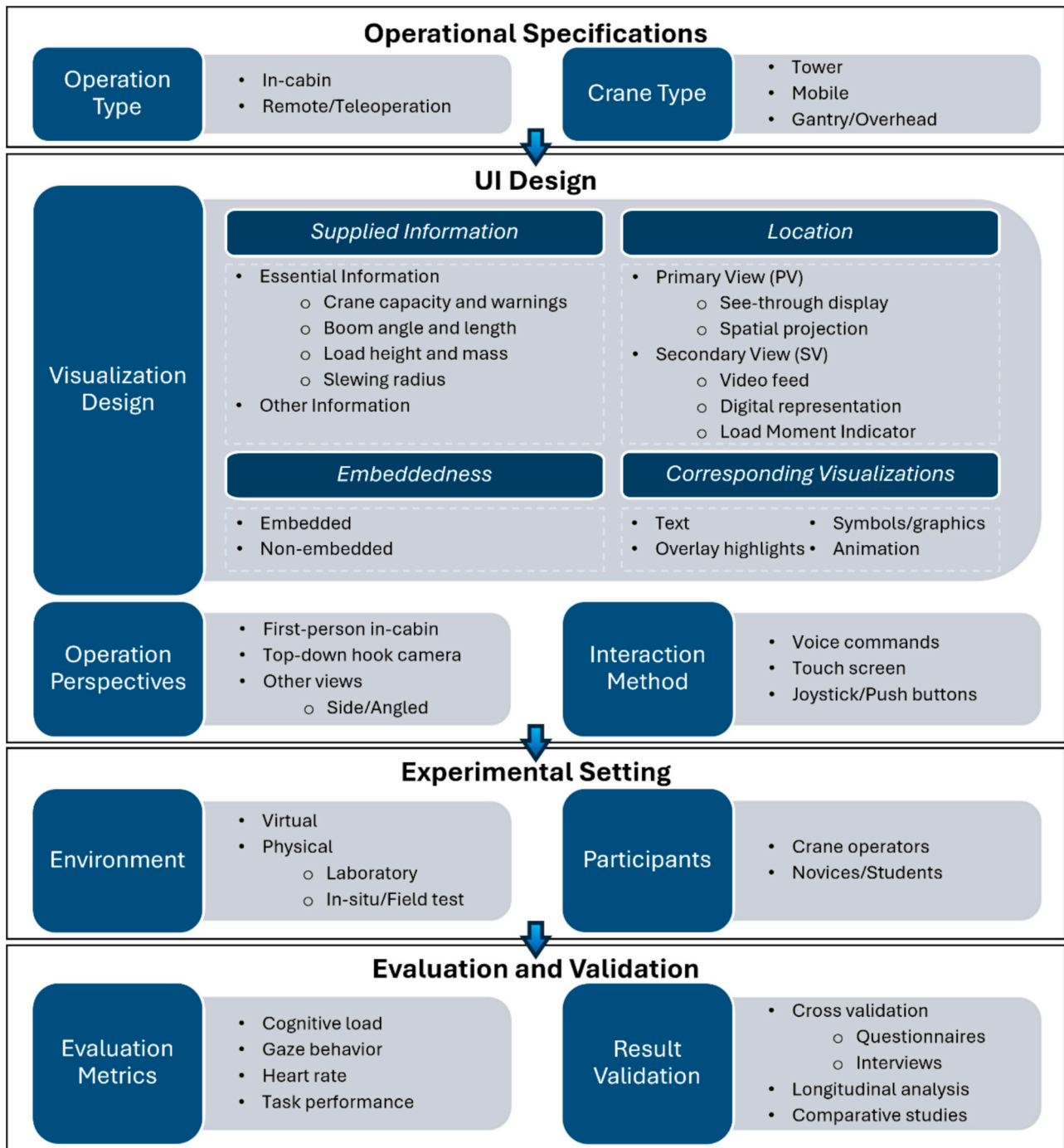


Fig. 6. Decision framework for the development of real-time crane assistant user-interfaces.

meet diverse scenario types and the evolving needs of the field.

4.2. Future research directions

4.2.1. Effectiveness of individual visualizations

One of the most significant gaps in current research is the limited focus on evaluating individual visualizations within crane UIs. Most studies concentrate on assessing the overall effectiveness of UIs rather than isolating and testing the visual elements that constitute them [53,54]. This generalization leaves open questions about which specific visualizations are the most effective for displaying critical information, particularly embedded visualizations. Studies that explore how embedded modes of information presentation compare to traditional

non-embedded approaches could offer valuable insights into improving operator performance and reducing cognitive load during crane operations.

Additionally, current research lacks understanding in two visualization areas: visualizations for error resolution and visualizations that aid depth perception. Visualizations for error resolution are essential for guiding operators out of hazardous events, providing clear and actionable feedback to prevent accidents. Currently there is an overreliance on text descriptions which may not be as effective as more intuitive visualizations in high-stress hazardous situations. Also, visualizations that enhance depth perception within the PV are also lacking. Further exploration of ideas similar to the hidden object highlights in [49,63] could potentially reduce the reliance on additional view angles,



Fig. 7. High-fidelity virtual crane environment rendered using Unity's High-Definition Render Pipeline (HDRP) with different weather effects (left – Sunny, right – Rainy and Foggy).

allowing operators to focus on a single view.

4.2.2. Comprehensive UI configurations

Another area that requires further investigation is the inclusion of multiple types of visualizations within a single interface. Several current studies either feature a limited number of visualizations [26,49,63,67,68] or suggest adding more information to the interface [33,50,56]. However, research has not yet explored the implications of integrating a more comprehensive range of visualizations, particularly in terms of managing the balance between cognitive load and situational awareness. A critical question for future research is how much embedded information can be included in a UI before it becomes overwhelming for operators. Additionally, determining the optimal layout for these visualizations is another underexplored aspect. Effective visualization layouts should prioritize critical information relevant to the current task while concealing less important ones. For instance, considering the different stages of a lift (i.e., initial hoist, slewing, lowering) as well as the various conditions that may be present (e.g., electrical hazards, swing-over risks, severe weather, low visibility due to fog or lighting, etc.). Identifying the importance of individual visualizations at any given moment necessitates the continuous involvement of crane operators who are intimately familiar with lifting procedures. Thus, engaging these operators in an iterative co-design process will be essential for creating interfaces that display the right amount of critical information in a particular scenario. Moreover, studies should consider incorporating customization options that allow interfaces to adapt to the varying experience levels and personal preferences of operators [62].

4.2.3. Complex scenario testing in VR

Current studies may not fully represent the complexity of real-world crane operations. For instance, the benefits of virtual information may not be immediately apparent in simplified experimental settings [33]. Future research should aim to develop more intricate scenarios that reflect the demanding environments faced by crane operators, such as tight clearances, critical lifts near crane capacity, blind lifts, low-visibility operations, or scenarios involving multiple cranes. Particularly, high-fidelity virtual reality (VR) platforms present a valuable opportunity to recreate these challenging situations at high levels of immersion and realism without the risks associated with in-situ testing. Studies such as [69] have demonstrated that high-fidelity virtual tower crane scenarios are achievable. These scenarios pair 3D meshes, created using drone capture, with realistic weather and lighting effects in the Unity game engine using its High-Definition Render Pipeline (HDRP) (Fig. 7). Future research should focus on whether visualization designs remain effective when operators face higher cognitive loads associated with more complex lift scenarios. Additionally, these high-fidelity environments will be ideal to further investigate the unique operational challenges faced by operators in different crane types as well as digital representations in secondary views (SV) that have predominantly occurred in real-world settings. Conducting these studies in VR could

alleviate the technological challenges of in-situ testing and ensure that the final designs are both practical and effective in enhancing safety and efficiency across different crane operations.

4.2.4. Collaborative features for enhanced communication

While crane UIs are primarily designed to assist the operator, future research should explore how they can facilitate better communication between lift personnel, such as signallers. Collaborative features that allow crane operators and other lift personnel to view and interact together with visualizations in real time are largely absent from current studies. Integrating these features could further enhance safety and coordination during complex lifts, especially in scenarios where direct communication is difficult or visibility is poor. Furthermore, as summarized by [63], this could also reduce the reliance on verbal communication between personnel and potentially allow them to designate more attention to other elements within their working environment.

4.2.5. Visualizations for scheduling

Effective scheduling of crane operations is essential for optimizing workflow and minimizing downtime at construction sites. Despite its significance, there is limited research specifically focused on assisting operators with consecutive lifts. Embedded visualizations of scheduling information, derived from Gantt charts and overlaying data on associated loads, could significantly enhance scheduling efficiency. These visualizations would provide operators with clear insights into operational timelines and potential conflicts, enabling them to make more informed decisions. Researchers should pay careful attention to ensure that these visualizations clearly indicate the sequence of tasks and transitions between them, allowing operators to anticipate and prepare for each stage of the operation. By integrating real-time data with visual scheduling tools, operators could better coordinate tasks and improve overall site productivity.

4.2.6. AI applications in UI development

Artificial Intelligence (AI) is a broad term which encompasses various branches of technology such as machine learning, natural language processing (NLP), and computer vision. To a slight extent, NLP has seen some applications in voice activated UIs to adjust zoom and perspective angles on a digital representation of a crane environment [56,57]. However, the broader integration of AI in UI design for real-time operations still holds significant potential and should be leveraged to create adaptive interfaces that respond dynamically to operator needs. For instance, real-time processing of gaze behavior data can optimize the presentation of visualizations, ensuring they are easily perceived. This could involve dynamically displaying visual cues in the center of an operator's line of sight or dynamically adjusting visualization parameters like its size, color, or opacity based on factors like environmental lighting. Additionally, machine learning algorithms can identify patterns in user interactions and performance data to predict behavior. This would allow UIs to suggest real-time improvements to

operator performance or even aid in the development of more effective safety alerts during the design stage.

Noting current barriers to the adoption of AI based technology, researchers are encouraged to also look into the development of modular components that can be integrated into existing systems without overhauling the entire UI. This stepwise approach allows for gradual adoption and testing of new technologies. Overall, leveraging AI technologies can lead to adaptive interfaces that enhance safety and efficiency in real-time crane operations. Exploring the vast number of possibilities could lead to significant advancements in this field.

4.2.7. Transferability of design principles

The design principles developed for crane UIs could also be applied to other types of machinery used in construction operations, such as excavators. Despite the potential for cross-industry application, there is little research exploring how these principles could be adapted for different machines. Future studies should investigate whether the augmented information techniques used for cranes can be effectively transferred to other heavy machinery, as well as how the unique operational contexts of different machines might require adjustments to UI design.

5. Conclusions

This paper provided a comprehensive assessment of visual information systems for crane operators, offering new insights into their effectiveness through a systematic literature review and the creation of a tailored UI evaluation framework. By synthesizing key trends, technologies, and design principles, this paper contributes significantly to the understanding of how user interfaces (UIs) can improve crane operations. A primary outcome is the development of a comprehensive UI evaluation model that enables both researchers and practitioners to assess multiple dimensions of UI design, such as consistency, clarity, usability, and error prevention. This model serves as a robust tool for analyzing the effectiveness of UIs in real-time crane operations, advancing both academic research and industry practices. Furthermore, this paper presents a general decision-making framework to complement the UI evaluation model as a starting point in the design and evaluation of real-time crane operation UIs. One of the paper's most valuable contributions is the identification of gaps in current research, particularly the need to evaluate individual visualizations within crane UIs and to explore more comprehensive UI configurations.

While this paper offers important contributions, it acknowledges several limitations. Firstly, this review assumes that the included studies have adequately addressed concerns regarding data collection and processing, such as accuracy and latency, without independently verifying these aspects. Secondly, the UI evaluation model has not been empirically validated across a broad range of crane types or real-world scenarios, restricting its generalizability. Only 14 UIs were assessed, which may limit the scope of the findings. Additionally, the model focuses primarily on design-related aspects relevant to real-time operations but excludes other critical factors such as device setup, maintenance, data privacy, and documentation, all of which are essential for practical implementation. Furthermore, although the research emphasizes the importance of involving crane operators in the design process, a larger sample size and a broader range of operator expertise are necessary to ensure that findings are reflective of diverse operational needs and would enhance the practical relevance and usability of future UI designs.

The contributions of this paper to the field of human-computer interaction in industrial settings are substantial. By identifying critical gaps in current UI designs, this research paves the way for future studies that can focus on enhancing operator performance through better design practices. The implications for the industry are profound, as improved UIs can lead to safer and more efficient crane operations, ultimately reducing accidents and increasing productivity on construction sites. To

realize these benefits, future research must focus on practical next steps like the prioritization of the development of prototypes in high-fidelity virtual reality (VR) environments, with laboratory scale studies allowing for the initial development of physical prototypes with minimal risk. Additionally, future researchers should stay updated on technological, social, and economic barriers to industry adoption and involve industry professionals in the design process to incorporate their practical insights. On that note, this paper also recommends the development of modular components for AI elements that may face greater implementation resistance, promoting gradual integration. Furthermore, future research should also consider exploring the development of collaborative visualization for better communication among crane personnel, and scheduling visualizations which have seen little attention in existing studies. By bridging these gaps, future studies can ensure that UIs evolve to meet the growing demands of modern crane operations, fostering safer and more efficient workflows in increasingly complex construction environments.

CRediT authorship contribution statement

Jiantsen Goh: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Yihai Fang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization. **Barrett Ens:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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